CBSE CLASS 10 GEOGRAPHY

Chapter 5: Minerals and Energy Resources
Detailed Chapter Assessment & Practice Question Bank (CBSE Pattern)

SECTION A: Multiple Choice Questions (1 Mark Each)

Q1. Which variety of iron ore is considered the finest quality with up to 70% iron content and possesses excellent magnetic properties?

(a) Hematite
(b) Magnetite
(c) Siderite
(d) Limonite

Q2. In which type of rocks do minerals like coal, gypsum, potash, and sodium salt occur in beds or layers as a result of deposition, accumulation, and concentration?

(a) Igneous rocks
(b) Metamorphic rocks
(c) Sedimentary rocks
(d) Volcanic rocks

Q3. Koderma in Jharkhand is the leading producer of which of the following minerals?

(a) Bauxite
(b) Mica
(c) Iron Ore
(d) Copper

Q4. Read the following statements regarding Manganese and select the correct option:
I. It is primarily used in the manufacturing of steel and ferro-manganese alloy.
II. Nearly 10 kg of manganese is required to manufacture one tonne of steel.
III. Odisha is a major producer of manganese in India.

(a) Only I is correct
(b) Only I and II are correct
(c) Only II and III are correct
(d) All I, II, and III are correct

SECTION B: Very Short Answer Questions (2 Marks Each)

Q5. Define the term 'Mineral'. How do geologists classify minerals based on their commercial value?

Q6. Distinguish between ferrous and non-ferrous metallic minerals, giving one example of each.

Q7. What are 'Placer Deposits'? Give two examples of minerals found in such deposits.

SECTION C: Short Answer Type Questions (3 Marks Each)

Q8. Why is Mica considered an indispensable mineral in the electrical and electronic industries? Give three reasons.

Q9. 'Mining activity is often called a hazardous industry.' Justify this statement by highlighting its impacts on miners and the environment.

Q10. Differentiate between conventional and non-conventional sources of energy with suitable examples.

SECTION D: Long Answer Type Questions (5 Marks Each)

Q11. Explain the distribution and importance of coal in India. Highlight the characteristics of its four major types (Anthracite, Bituminous, Lignite, and Peat).

Q12. Why is there a pressing need to use non-conventional or renewable energy resources in India? Provide five well-supported arguments.

Q13. 'Conservation of mineral resources is essential for sustainable development.' Examine the statement and suggest four practical measures to conserve minerals.

SECTION E: Case-Based/Source-Based Questions (4 Marks)

Read the source given below and answer the questions that follow:
Natural gas is an important clean energy resource found in association with or without petroleum. It is used as a source of energy as well as an industrial raw material in the petrochemical industry. Natural gas is considered an environment-friendly fuel because of low carbon dioxide emissions and is, therefore, the fuel for the present century. Large reserves of natural gas have been discovered in the Krishna-Godavari basin. The 1,700 km long Hazira-Vijaipur-Jagdishpur (HVJ) cross-country gas pipeline links Mumbai High and Bassein with the fertilizer, power, and industrial complexes in western and northern India.

Questions:

1. Why is natural gas considered an environment-friendly fuel? (1 Mark)

2. Name the major cross-country pipeline that connects gas fields of western India to northern markets. (1 Mark)

3. Highlight the economic significance of natural gas as an energy resource and an industrial raw material. (2 Marks)

SECTION F: Map Skill Based Questions (3 Marks)

Q14. On an outline political map of India, locate and label the following with appropriate symbols:

1. **Mayurbhanj** - Iron ore mine
2. **Digboi** - Oil field
3. **Tarapur** - Nuclear power plant